Lab: Derivation Exercise — Policy Evaluation and Sophisticated Inference

Objective

Derive and compute Expected Free Energy for multi-step policies, extending to sophisticated inference with recursive belief updating.

Part 1: Two-Step Policy Evaluation

Goal: Compute EFE for all policies in a simple POMDP.

Setup: 2 states, 2 observations, 2 actions, planning horizon T = 2.

- A matrix: A = [[0.9, 0.1], [0.1, 0.9]]
- B matrix: B_1 = [[0.8, 0.2], [0.2, 0.8]] (action 1), B_2 = [[0.3, 0.7], [0.7, 0.3]] (action 2)
- C vector: C = [ln 2, ln(1/2)] (prefer observation 1)
- Initial belief: $q(s_1)$ = [0.5, 0.5]

There are K^T = 2^2 = 4 policies: $\pi_1=(a_1,a_1)$, $\pi_2=(a_1,a_2)$, $\pi_3=(a_2,a_1)$, $\pi_4=(a_2,a_2)$.

Task:

1. For each policy, compute $q(s_2 \mid \pi) = B(\pi_1) \cdot q(s_1)$ and $q(s_3 \mid \pi) = B(\pi_2) \cdot q(s_2 \mid \pi)$
2. Compute predicted observations $q(o\_\tau \mid \pi) = A \cdot q(s\_\tau \mid \pi)$ for each timestep
3. Compute the EFE $G(\pi) = \sum\_\tau$ [pragmatic_τ + epistemic_τ] for each policy
4. Select the best policy (lowest G)

Write your response here

Part 2: Sophisticated Inference

Goal: Extend Part 1 with future belief updates.

Task:

1. For the best policy from Part 1, simulate what happens at timestep 2: the agent observes o_2, then updates its belief using Bayes' rule: $q(s_2 \mid o_2, \pi) \propto A(o_2, :) \cdot q(s_2 \mid \pi)$
2. For each possible observation o_2, compute the updated belief $q(s_2 \mid o_2, \pi)$
3. Evaluate the EFE of the second action using these updated beliefs instead of the prior beliefs
4. Compare the sophisticated EFE with the basic EFE from Part 1. How do they differ?

Write your response here

Part 3: Hierarchical POMDP

Goal: Construct a two-level hierarchical POMDP.

Setup:

- Level 1 (fast): 3 states, 2 observations, 2 actions, runs for $T_1 = 3$ steps per Level 2 step
- Level 2 (slow): 2 states (contexts), determines the initial state distribution and preferences for Level 1

Task:

1. Write the generative model for each level
2. Show how Level 2's policy selection sets the D vector (initial state prior) and C vector (preferences) for Level 1
3. Derive the total free energy $F\_total = F_1 + F_2$
4. Explain how optimizing at Level 2 selects which sub-task to pursue, while Level 1 handles moment-to-moment execution

Write your response here

Part 4: Computational Complexity Analysis

Goal: Analyze the scalability of planning.

Task:

1. For a flat POMDP with K actions and planning horizon T, count the number of policies: K^T
2. For K = 4, compute the number of policies for T = 1, 2, 3, 5, 10
3. Propose a pruning strategy based on Expected Free Energy thresholds: at each timestep, keep only the top-N policies
4. Analyze the complexity reduction: from K^T to N·K per timestep
5. Discuss: how does habit formation (defaulting to well-learned policies) further reduce computational demands?

Write your response here

Part 5: Synthesis

In 200 words, explain how the three planning levels (basic EFE, sophisticated inference, hierarchical POMDPs) provide increasingly powerful but computationally costly planning — and how biological brains manage this trade-off through habits, sub-goals, and selective deliberation.

Write your response here

Lab Summary

Part	Skill Developed	Mathematical Result
1	Numerical computation	Full EFE evaluation for all policies
2	Recursive reasoning	Sophisticated inference with future belief updates
3	Hierarchical modeling	Two-level POMDP with context-dependent sub-tasks
4	Complexity analysis	Scalability and pruning strategies
5	Conceptual integration	Biological-computational planning trade-offs